2014 7th Cairo International Biomedical Engineering Conference
Cairo, Egypt, December 11-13, 2014

Compressed-sensing-based three-dimensional image reconstruction algorithm for C-arm vascular imaging

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***Abstract*—X-ray C-arm is an important imaging tool in interventional surgery, road-mapping and radiation therapy. It provides accurate description of vascular anatomy and therapy end point. The C-arm scanner produces two-dimensional (2D) x-ray projection data obtained with flat-panel detector by rotating the source around the patient. The number of 2D projections acquired is several hundreds, which results in significant amount of radiation dose. Unlike the conventional fluoroscopic imaging, three-dimensional (3D) C-arm computed tomography (CT) provides more accurate cross-sectional images which are valuable for therapy planning, guidance and evaluation in interventional radiology. However, 3D vascular imaging using the conventional C-arm fluoroscopy is a challenging task. First, the rotation orbit of the C-arm gantry is usually limited to a range less than those of CT scanners. Second, in several commercial models (including the one of consideration in this study), the x-ray source and detector are shifted from the gantry isocenter to enlarge the scanner field-of-view (FOV), which is so-called the offset scan. Finally, it is difficult to acquire sufficient projection views required for stable 3D reconstruction using manually controlled gantry motion. Inspired by the theory of compressed sensing, we developed an image reconstruction algorithm for the conventional angiography C-arm scanners. The main challenge in this image reconstruction problem is the projection data limitations. We consider a small number of views (less than 10 views) acquired from a short orbit with the offset scan geometry. The proposed method is developed using the alternating direction method of multipliers (ADMM) and results obtained from simulated data and real data are encouraging. The proposed method can significantly contribute to the reduction of patient dose and provides a framework to generate 3D vascular images using the conventional C-arm scanners.**

***Keywords*—Image reconstruction; computed tomography; 3D C-arm CT; ADMM**

I. INTRODUCTION

C-arm angiography is an effective imaging tool for many different types of physicians such as cardiologist, radiologist, surgeons, urologist, orthopedic surgeries and neurovascular applications. It is an important tool in interventional surgery, radiation therapy, road mapping and guidance [1]. Image-guided interventional radiology aims to provide physician with anatomical and physiological structure of the patient organs during minimal invasive surgery procedures. In typical protocols for imaging blood vessels, the patient is injected with contrast agent, and the acquired two-dimensional (2D) images are processed through digital subtraction angiography (DSA) approach. In DSA, image acquired in prior to the injection of the contrast agent is subtracted from contrast-based acquired images to remove surrounding complicated structures [2]. Sample images acquired for kidney tumor vessels using the DSA approach are shown in Fig. 1. The common challenge in this procedure is that 2D images are not enough to provide required three-dimensional (3D) information of vessel anatomy due to the complicated structure and difficulties in identifying the overlapped vessels (Fig. 1 (c)). In clinical procedures, physicians usually change the image acquire angle by rotation the C-arm gantry to get different-view images (see Fig. 1 (d)) and mentally process these images to understand the 3D vessel structure. This procedure is known to increase the x-ray dose given to patient in addition to the contrast agent as well. Also, it requires special clinical experience to be processed successfully. In some applications, the C-arm is placed close to a computed tomography (CT) scanner in order to perform a CT scan simultaneously to observe the 3D structure of vessels using CT images. However, this approach also increases the patient dose and patient is required to be moved from the surgery table to the scanner table which is inconvenient in

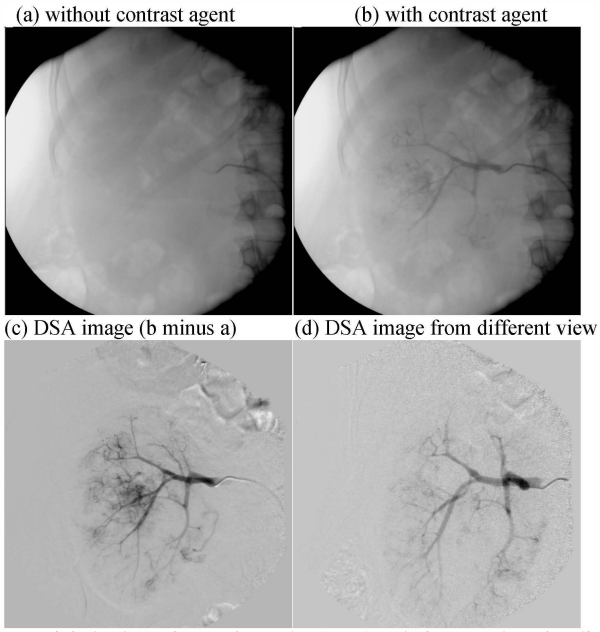

Fig. 1. Digital subtraction angiography (DSA) technique used to visualize the arteries of a kidney tumor (*image courtesy of one of the authors*).

This work is partially supported by the Applied Scientific Research Projects (ASRP) program, Ministry of Higher Education, Egypt under Grant number ASRP1-008-SUE.

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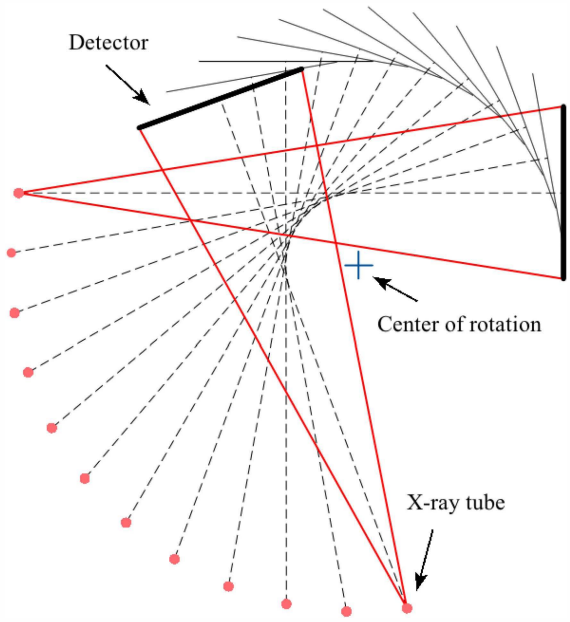

Fig. 2. Sketch of imaging configurations of the Ziehm Vision R C-arm scanner with orbit rotation maximum of 115º and with offset scan geometry. The red points and dashed lines illustrates the positions x-ray source and lines from the source to detector, respectively, corresponding to equally spaced 12 projection views with angular spacing of 10º.

some cases. Three-dimensional C-arm CT uses 2D x-ray projection data obtained with a flat-panel detector by rotating the C-arm around the patient to generate CT-like images. The number of required projections is approximately several hundred of views. Modern multi-axis C-arm systems (such as Siemens Artis Zeego) can provide 3D images by recording several projections around the patient and perform tomographic image reconstruction. However, these advanced systems are expensive compared to other standard C-arm scanners [1].

In this study, we aim to use the conventional 2D C-arm scanners to provide 3D vascular images. The main module in the proposed framework is an image reconstruction algorithm from limited projection data. The problem of tomographic image reconstruction of sparse object from limited projection data has been studied in literature (e.g. see Refs. [3], [4]). The proposed approach is based on the theory of compressed sensing [5], [6]. The image reconstruction algorithm is developed to handle sparse objects such as blood vessels similar to the one shown in Fig. 1(c), where most of image pixels are zeros or close to zeros and a small number of image pixels have non-zeros intensities.

II. METHODS

A. System geometry and data acquisition

Data limitations of the imaging system under study consist of a combination of a few famous data limitations which arise from imaging setups of our C-arm scanner [7]-[9]. First, the scanner gantry is manually operated, so we can acquire only a small number of projection views. Measuring only a small number of projections has great benefit to reduce the accumulated patient dose. However, the corresponding image

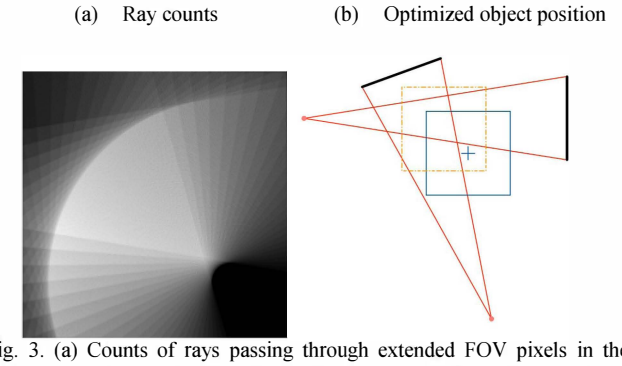

Fig. 3. (a) Counts of rays passing through extended FOV pixels in the offset scan geometry shown in Fig. 2 computed using 24 projection views. (b) The optimized position of the imaged object in dashed line which is shifted from the original center of rotation by approximately 92 mm in both x and y coordinate directions.

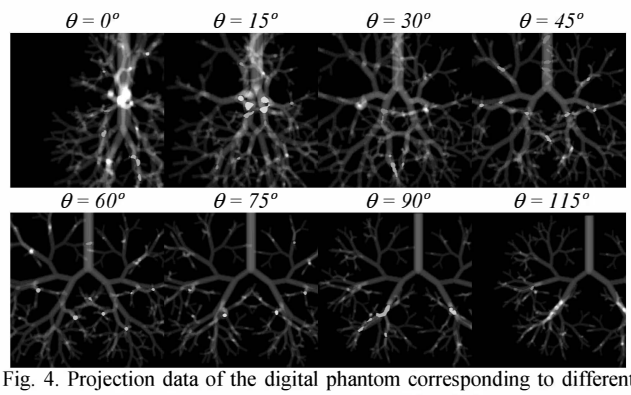

Fig. 4. Projection data of the digital phantom corresponding to different view angles θ. In all views, some part of the object is located outside the FOV.

reconstruction problem becomes unstable and achieving exact reconstruction is difficult. Second, the gantry rotation orbit is limited to a range of angles less than the theoretically required one for stable CT reconstruction. The imaging system used in this work has a rotation range limited to 115º, which is below the normal angular range used in CT reconstruction. Third, we consider the offset scan geometry, where the x-ray source and detector are shifted from the central line that passes through the gantry isocenter. This setup is common in C-arm scanners to increase the scanner FOV in 2D fluoroscopy. However, it is an uncommon setup for 3D image reconstruction. It is known that, in the offset scan geometry, the common region which can be viewed in all projection data is rather small and thus most of projection data suffer from data truncation. To illustrate the data limitation problems of the imaging geometry under study, we show the imaging configuration of a commercial C-arm scanner used in this work in Fig. 2. The combination of data limitation problems described above produces a challenging imaging system for which it is difficult to develop an accurate image reconstruction algorithm with stable solution. To reduce the effect of data truncation due to the offset scan geometry, we study the extended FOV to determine the area in which the number of projection rays passing through the FOV is maximized, thus the effect of data truncation is minimized. This is simply achieved by counting the number of projection rays passing through each image pixel. Results are shown in

Fig. 3(a). It is clear from these results that the object should not be located in the gantry center of rotation. However, it should be shifted up and left with a value of approximately 92 *mm* to minimize the effect of truncation. The suggested object position is illustrated in Fig. 3(b).

B. Image reconstruction algorithm

In x-ray CT, the image reconstruction problem can be formulated as solving the following system of linear equations

$$Ax = b \tag{1}$$

where $x = (x_1, \ldots, x_n)^T$ is a vector representing the object image with n pixels, $b = (b_1, \ldots, b_m)^T$ is a vector consisting of m measured projection data and A is the $m \times n$ system matrix that represent system geometries. In the application of vascular imaging, vector x is known to be sparse (See Fig. 1 (c)). A potential method to solve the linear system in Eq. (1) is the equality constrained ℓ_1-norm minimization problem, known as the basis pursuit

$$\text{minimize } \|x\|_1 \text{ subject to } Ax = b \tag{2}$$

The solution of problem in Eq. (2) can be achieved by using several approaches such as the alternating direction method of multipliers (ADMM) [10]. In ADMM, basis pursuit problem in Eq. (2) can be reformulated as:

$$\text{minimize } f(x) + \|u\|_1 \text{ subject to } x - u = 0 \tag{3}$$

where f is the indicator function of the affine subspace $\{x \in \mathcal{R} \mid Ax = b\}$. The ADMM algorithm is performed through the following steps.

$$x^{k+1} = \Pi\,(u^k - \lambda^k) \tag{4}$$

$$u^{k+1} = \text{SoftThres}(x^{k+1} + \rho^{-1}\lambda^k), \tag{5}$$

$$\lambda^{k+1} = \lambda^k + x^{k+1} - u^{k+1} \tag{6}$$

where ρ is the penalty parameter used as the step size, k is the iteration number, Π is the projection onto $\{x \in \mathcal{R} | Ax = b\}$ and SoftThres is the soft thresholding operator defined as.

$$\text{SoftThres}(t, a) = \begin{cases} a - t & a > t \\ a + t & a < -t \\ 0 & |a| \le t \end{cases} \tag{7}$$

The ADMM algorithm consists of three main steps. The first step is the enforcement of data fidelity term as in Eq. (4). The second step is the soft-thresholding operation (Eq. (5)) and the last step is the multiplier update step as in Eq. (6). In results the presented in this paper, Eq. (4) is implemented using the well-known Algebraic Reconstruction Technique (ART) algorithm [12]. The interesting behavior of this algorithm is that, in practical implementation, it is enough to use a single inner iteration of data fidelity enforcement in each iteration k, to obtain a high-quality image. The theoretical explanation of this behavior is yet unclear. The proposed iterative reconstruction method is of potentially useful for the target application. It provides quick, effective and easy-to-implement reconstruction method. Its computation cost is almost the same as those of several conventional methods belonging to the Projections Onto Convex Sets (POCS) family.

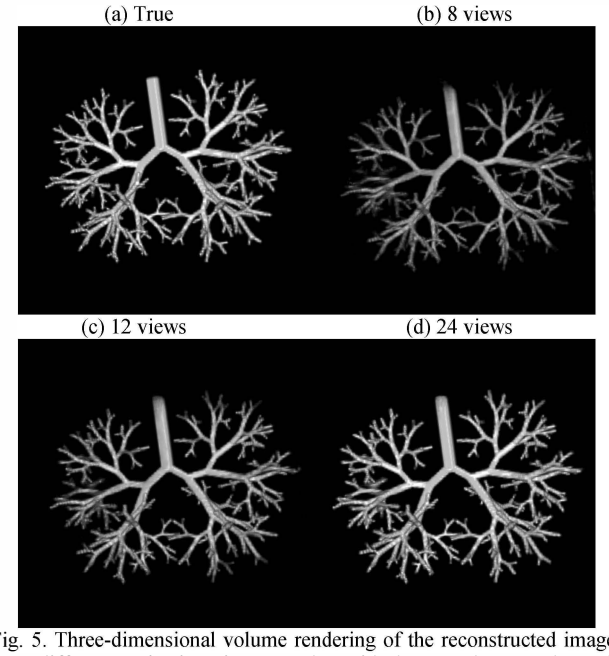

Fig. 5. Three-dimensional volume rendering of the reconstructed images from different projection views together with the true phantom shown in (a).

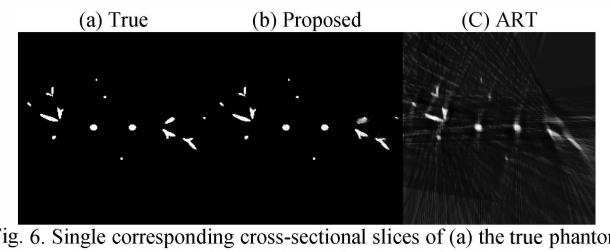

Fig. 6. Single corresponding cross-sectional slices of (a) the true phantom and reconstructed images from 24 views using (b) the proposed method and (c) the ART algorithm.

III. Experimental Studies

A. Simulated data

A computer simulation study was performed to evaluate the proposed method. A digital phantom (FORBILD lung phantom [11]) representing a vessel-like sparse structure was used in this study. The phantom volume is 256^3 pixels and the cone-beam projection data are acquired at 8, 12 and 24 views with rotation angle spacing of 15º, 10 º and 5 º, respectively, over gantry rotation arc of 115 º. The scanner geometry parameters, shown in Table I, was set to real parameters of the Ziehm Vision R C-arm scanner (Ziehm Imaging GmbH, Nürnberg, Germany). Due to the scanner geometry settings, the scanner FOV, which is visible in all view angles is small and, consequently, all projection views suffer from data truncation. In Fig. 4, we show the simulated 2D detector measurements in different view angles θ. The data incompleteness in each projection view produces an additional challenge in image reconstruction in additional to a small number of projection views.

Three-dimensional images reconstructed from 8, 12 and 24 projection views together with the true object are shown in Fig.5 and the corresponding transverse cross-sectional slices are shown in Fig. 6. From these results, it can be observed that

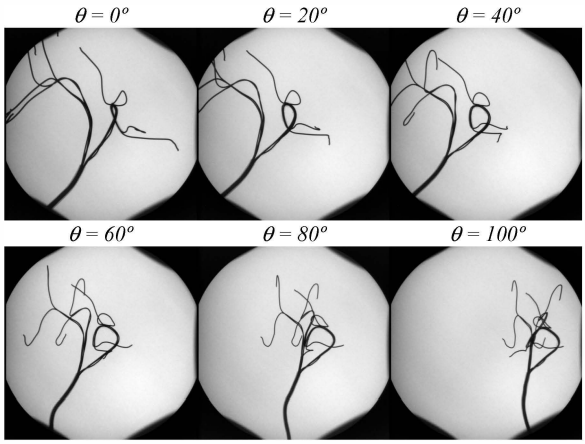

Fig. 7. Real projection data acquired from metal wire phantom, shown in Fig. 8(a), measured by the C-arm detector in different view angles θ.

the proposed method can reconstruct an accurate image even with eight projection views as shown in Fig. 4 (b).

B. Real data

To evaluate the proposed method, we have carried out a real data experiment. A metal wire was used to simulate the human segmental hepatic arterial system imaged with DSA approach. The metal wire was imaged using Ziehm Vision R C-arm scanner installed in our hospital. The scanner gantry was adjusted manually to acquire 24 projection views with angle spacing of 5º. Examples of acquired detector measurements extracted in DICOM image format are shown in Fig. 7. Image reconstruction was performed using $\rho=1$ and 10 iterations and volume rendering of reconstructed image along with photograph of the wire object are shown in Fig. 8. As it is difficult to show a volume rendering of the ART reconstruction with strong artifacts, we have shown a cross-sectional orthogonal views of the reconstruction using the proposed method and the ART algorithm in Fig. 9. It is clear that the reconstruction using the proposed method is superior to conventional ART method in terms of image quality and suppression of streak artifacts.

IV. Conclusion

We proposed an iterative image reconstruction algorithm for 3D vascular imaging using C-arm scanner. The developed algorithm was designed to overcome data limitation challenges in this imaging modality including limited views, short orbit and offset scan geometries. This framework aims to provide an effective 3D imaging using conventional C-arm scanners, which are originally designed to generate 2D fluoroscopy images. This work has several benefits: (1) it provides 3D images using conventional inexpensive equipment, (2) there is a great potential to reduce patient dose, interventional procedure time, injected contrast agent and (3) it is expected to minimize the failure rate of several interventional procedures by providing the physician with clear 3D structure of the vessels. This is an initial study and preliminary results using simulated and real data are encouraging. Future studies will include implementation of automatic data acquisition system and more accurate geometry calibration.

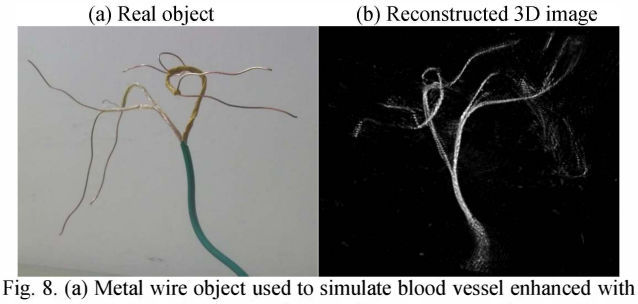

Fig. 8. (a) Metal wire object used to simulate blood vessel enhanced with contrast agent. (b) Three-dimensional volume rendering of reconstructed image of the metal wire from 24 projection views.

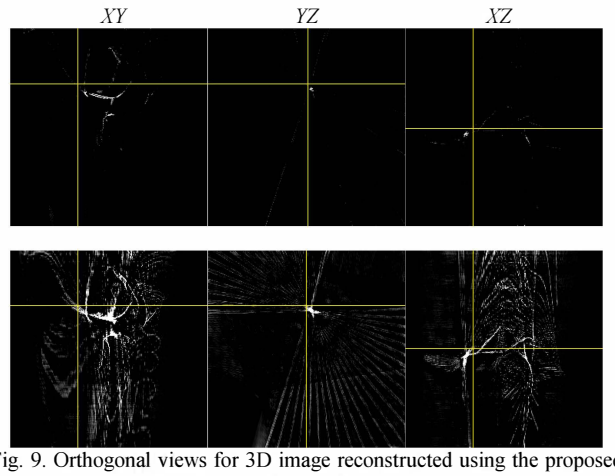

Fig. 9. Orthogonal views for 3D image reconstructed using the proposed method (top) and the ART algorithm (below).

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